

Experiment No. 23 (Mixture No. 1)

Date: / /

Aim : Analyse two basic (cation) radicals qualitatively from given inorganic mixture.

Apparatus : Test tubes, test tube holder, test tube stand, filter paper etc.

A. Preliminary tests:

Test	Observation	Inference
Color	white	Sr^{2+} , Ba^{2+} may be present
Nature	crystalline	water soluble salts present.

B. Dry Tests for Basic radicals (Test tube must be dry for this test):

Test	Observation	Inference
1. Heating in a dry test tube:- Take a small quantity of the mixture a clean and in dry test tube and heat it strongly.	i) cracking noise ii) white residue which glows on heating	crystalline salt like $Pb(NO_3)_2$, $NaCl$, KBr may be present. Ba^{2+} , Sr^{2+} , Ca^{2+} , Mg^{2+} salt may be present
2. Charcoal Cavity Test: Mixture + Na_2CO_3 solid in 1:2 proportion placed in fresh charcoal cavity, moisten with a drop of water. Heat it with blow pipe in a reducing (yellow) flame	white residue	Al^{3+} , Ba^{2+} , Ca^{2+} , Mg^{2+} salts may be present
3. NaOH Test: Mixture + NaOH solution & heat. Hold moist turmeric paper near the mouth of the test tube	moist fuming	NH_4^+ absent
4. Flame Test: Prepare a paste of the given mixture with conc. HCl on a watch glass. Make a small loop at the end of the platinum wire & dip it in the mixture or use glass rod. Heat it on oxidising flame (Blue) observe the colour change of the flame.	Pale green or Apple green	Ba^{2+} may be present

C. Preparation of original solution (O.S)

Take a small quantity of mixture in a beaker add 20 mL of distilled water, stir with glass rod to dissolve the mixture. If mixture does not dissolve completely then warm it to dissolve. Clear solution is obtained, which is used as a O.S for further tests.

1. Analysis of Group zero (NH_4^+)

Test	Observation	Inference
1. O.S. + NaOH solution + Heat, test with moist turmeric paper.	No evolution of NH_3 gas	Group zero is absent.

2. O.S. + NaOH solution + Heat, Bring a glass rod dipped in conc. HCl near the mouth of the test tube.	No done white forms of NH_4Cl	NH_4^+ is absent.
C.T. for NH_4^+ : 1. O.S. + Nessler's reagent in excess	—	—
2. O.S. + Picric Acid (2, 4, 6 trinitro phenol)	—	—

Detection of group (if group zero is absent, two groups must be detected following test)

Test	Observation	Inference
1. O.S. + dil HCl	No white ppt	Group I absent
2. O.S./Filtrate + dil HCl (heat) + H_2S gas or water	No black ppt	Group II absent
3. O.S./Filtrate (Remove H_2S) + NH_4Cl (equal) + NH_4OH (till alkaline to litmus)	white gelatinous ppt.	Group III (Al^{3+} , Fe^{3+}) present.
4. O.S./Filtrate + NH_4Cl (equal) + NH_4OH (till alkaline to litmus) + H_2S gas or water	No ppt	Group IV absent
5. O.S./Filtrate (Remove H_2S) + NH_4Cl (equal) + NH_4OH (till alkaline to litmus) + $(\text{NH}_4)_2\text{CO}_3$	white ppt	Group V Ca^{2+} , Ba^{2+} , Sr^{2+} present
6. O.S./Filtrate + NH_4Cl (equal) + NH_4OH (till alkaline to litmus) + Na_2HPO_4	—	—

2. Analysis of first detected group Group III (Al^{3+} , Fe^{3+})

Test	Observations	Inference
O.S. + NH_4Cl + NH_4OH	white gelabrous ppt	Al^{3+} present
—	—	—
—	—	—

3. C.T. for first detected radical Al^{3+}

Test	Observations	Inference
R.S + few drops of 1. NaOH and warm	white gelatinous ppt	Al^{3+} Confirmed
2. R.S + Blue litmus Sol ⁿ + NH_4OH drop wise	Blue floating mass in colourless sol ⁿ	Al^{3+} confirmed

2. Analysis of second detected group

Test	Observations	Inference
Filtrate + NH_4Cl	white ppt	Ca^{2+} , Sr^{2+} , Br^{2+} may be present
1. + NH_4OH + $(NH_4)_2CO_3$		
filtrate + NH_4Cl + 2. NH_4OH + $(NH_4)_2CO_3$	Yellow ppt	Br^{2+} is present
3. —	—	—

3. C.T. for second detected radical Ba^{2+}

Test	Observations	Inference
above acetate 1. sol ⁿ + dil H_2SO_4	white ppt.	Ba^{2+} confirmed
above solution 2. + $(NH_4)_2 CO_3$	white ppt	Ba^{2+} confirmed

Result:-

The given inorganic mixture no. 1 contains following two cations (Basic Radicals)

- Al^{3+} (Name of cation) Aluminium cation
- Ba^{2+} (Name of cation) Barium cation

Remark and sign of teacher: _____